

LogInsight Analyzer

Algorithm-Driven Log Analysis and Incident Investigation

DATA STRUCTURES AND ALGORITHMS-3 · 25CS2103E · PBL

Koneru Lakshmaiah Education Foundation · 2026-2027, Odd Semester

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1. Problem statement

During an outage the evidence required to explain what happened is spread across services, arrives interleaved and out of order, and is queried far faster than a human can read it. Substring search over the corpus is $O(nm)$ and rescans on every attempt, so the engineer never learns which algorithm actually ran or what it measured.

LogInsight applies classical data structures and algorithms to that work — multi-pattern and exact-substring search, bounded edit distance, streaming window aggregation and dependency-graph traversal — and presents the returned events, aggregates and algorithm evidence together with their source and scope.

2. Architecture

The application is three layers with one rule: telemetry is server-owned. The Spring Boot backend generates deterministic scenarios, computes product aggregates, executes the algorithm engines and manages process-local incident state. The React frontend renders API responses with their scope and provenance, and claims algorithm execution evidence only where a real endpoint produced it.

Transport is HTTP REST plus three Server-Sent Event streams that are deliberately kept distinct: the generated simulation, the bounded dataset replay, and recorded trace replay. There is no WebSocket client and no external log collector.

3. Data flow

A dataset is loaded from sample-data, uploaded, or generated by the backend.

DatasetService installs it as one immutable in-memory current dataset; readers take a consistent snapshot.

Analytics, incident detection, pattern extraction and search all derive from that dataset.

The frontend requests derived values and never recomputes backend analysis.

Mutating the dataset invalidates every cached view through a shared event.

4. Algorithms and their actual roles

The catalogue distinguishes algorithms exposed through an execution endpoint, algorithms that additionally emit a trace, and entries that exist as lab implementations. A wider repertoire — bitmask TSP, Needleman-Wunsch, Dinic flow, vertex-cover kernelization, parallel prefix scan and Miller-Rabin — is exposed as runnable lab entries rather than presented as product capability.

5. 2D and 3D visualisation

The service topology renders as an accessible 2D SVG by default, with a paired service and edge list so the same data is available without WebGL. An optional Three.js view is code-split and loaded on demand.

The WebGL scene is rebuilt only when the graph genuinely changes, not on every render.

Teardown disposes geometries, materials and controls, stops the animation loop, and explicitly releases the browser WebGL context, which renderer disposal alone does not do.

Context loss surfaces a usable 2D fallback; context restoration rebuilds the scene automatically.

Switching modes preserves the selected service and every active filter.

Node size follows observed event counts and health follows error-rate bands; neither implies causation.

6. Error handling and data honesty

Every dataset-backed request returns a structured error envelope. Validation and malformed input map to 400, a missing dataset to 404, and unexpected failures to a sanitised 500 that never carries a stack trace. When no dataset is loaded the interface shows onboarding rather than zeroed metrics, and unmeasured values render as an em dash instead of being coerced to zero.

4.1 Algorithm catalogue

Algorithm	Complexity	Role in the product
KMP	$O(n + m)$	Linear exact substring search; no backtracking.
Aho-Corasick	$O(n)$	Every error signature matched in one left-to-right pass.
Levenshtein / Damerau	$O(nm)$	Edit-distance DP ranks a 'did you mean' suggestion.
BFS over dependency graph	$O(V + E)$	Blast radius reachable downstream of the failing service.
Fixed-window detection	$O(n + W)$	Bucketed, baseline-relative threshold, merged into one window.
Suffix array + Kasai LCP	$O(n \log n)$	Traceable phrase search over the corpus.

7. Testing methodology and results

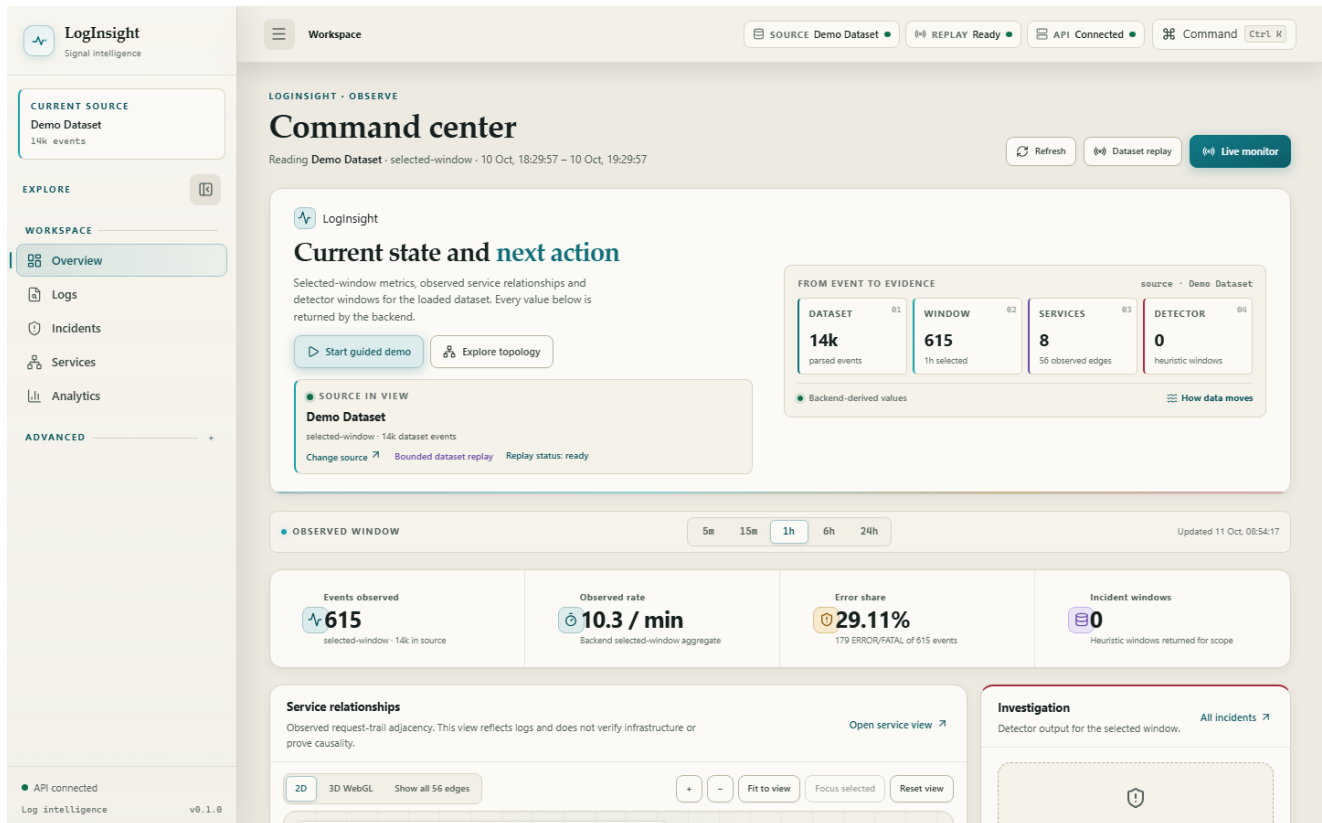
The browser suite covers direct navigation to every documented route, route aliases, dataset loading and replacement, search and pagination, empty and failing states, repeated 2D/3D mode switching, the WebGL fallback, the guided walkthrough, reduced motion, console and failed network requests, and responsive overflow at six viewport sizes.

Check	Result	Command
Backend tests	910	mvnw -B verify — 0 failures, 0 errors, 0 skipped
Frontend unit tests	72	vitest run — 72 across 20 files, single aggregate run
Browser end-to-end	18	Playwright — all routes, 2D/3D round trip, reduced motion
Dependency audit	0	npm audit — no vulnerabilities
Production build	pass	tsc && vite build

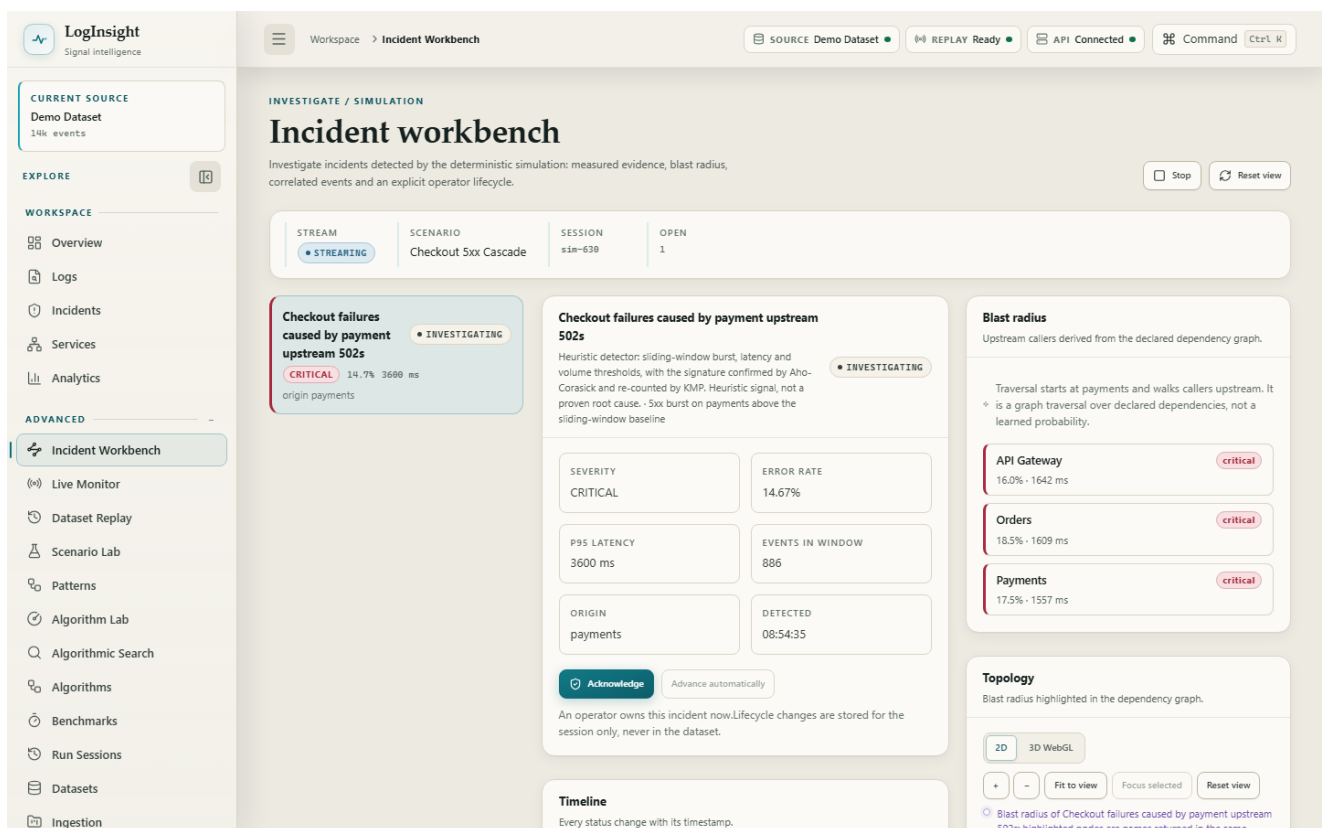
8. Performance

Measure	Before	After
First-load JS	before: shipped inside the entry chunk	after: 483 kB
Three.js	before: loaded with the application	after: deferred to the 3D topology

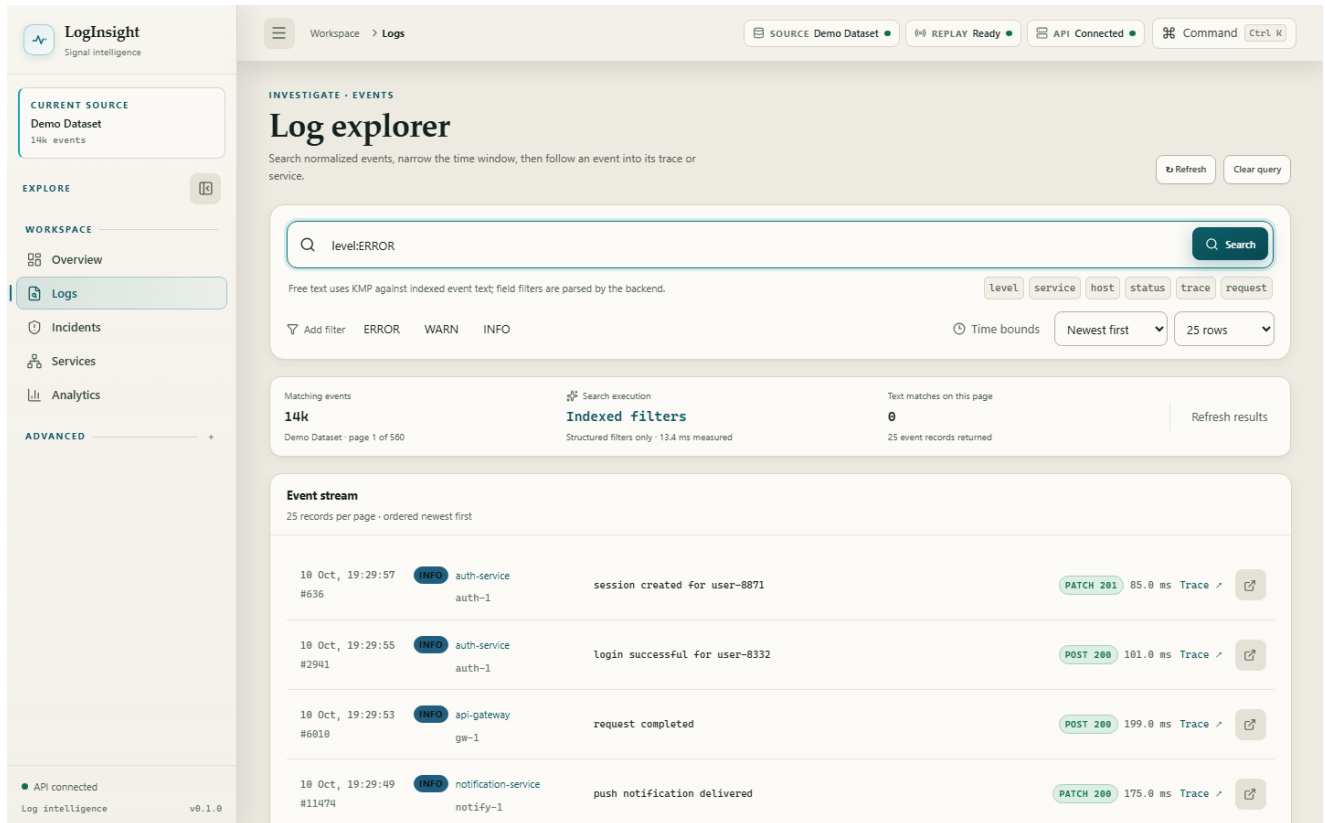
9. Screenshots from the final implementation



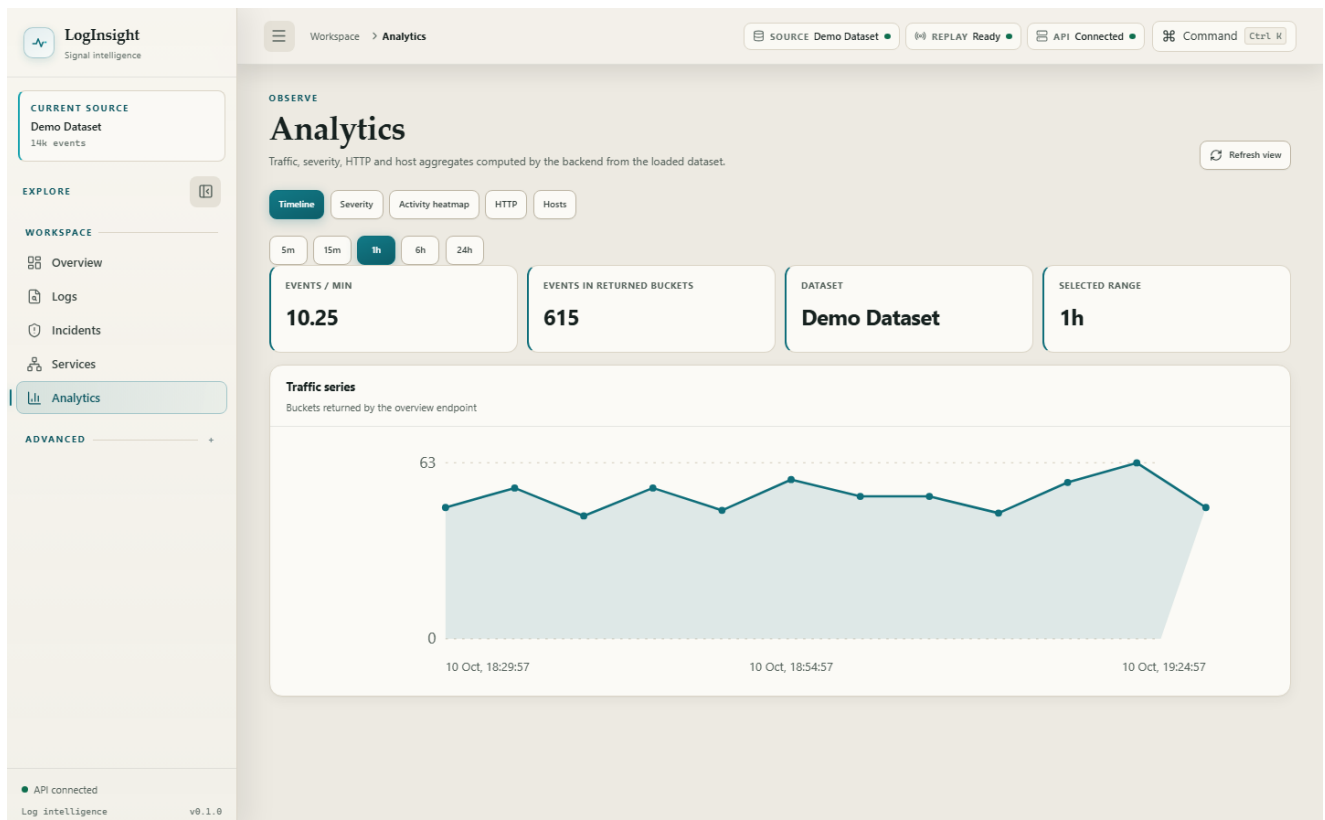
Command center with backend-derived selected-window values.



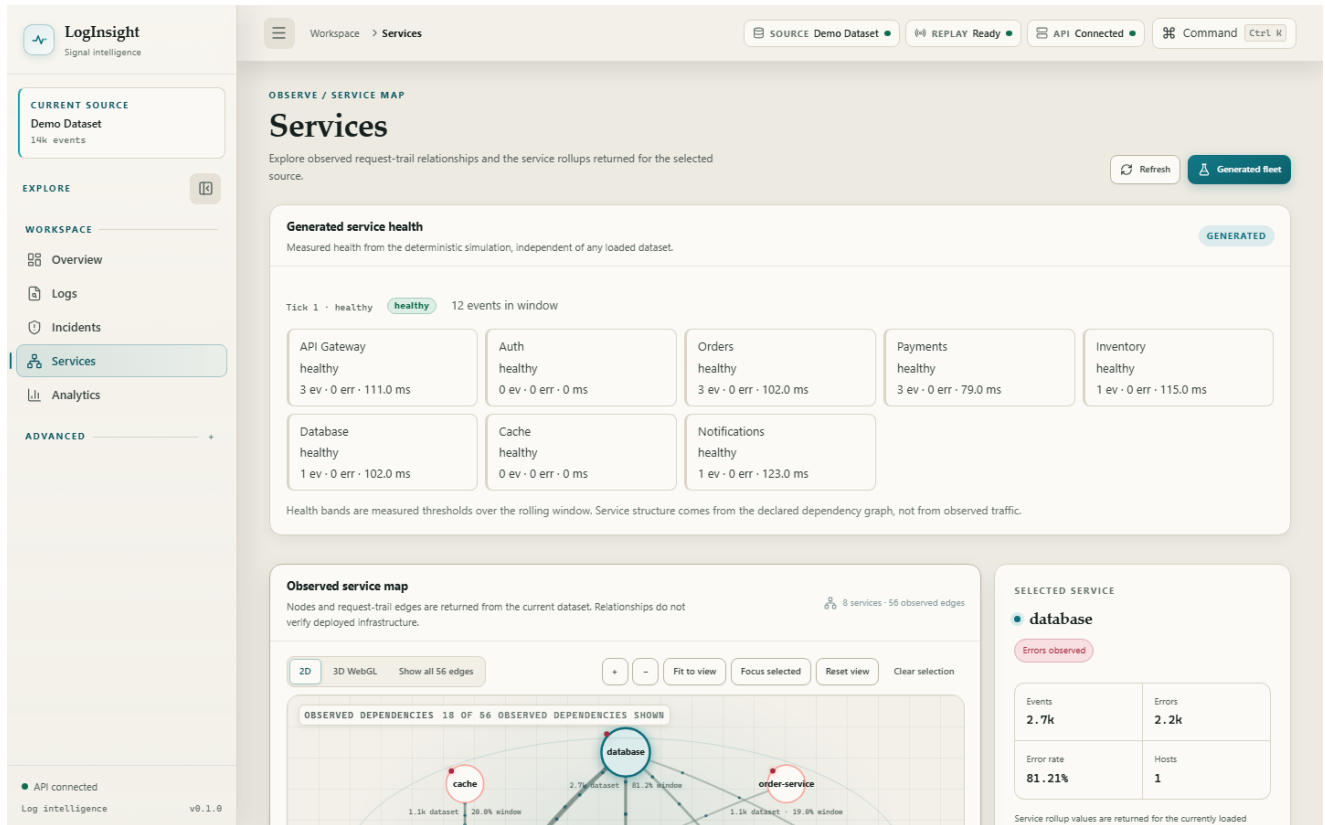
Incident workbench: identity, measured context, blast radius, lifecycle.



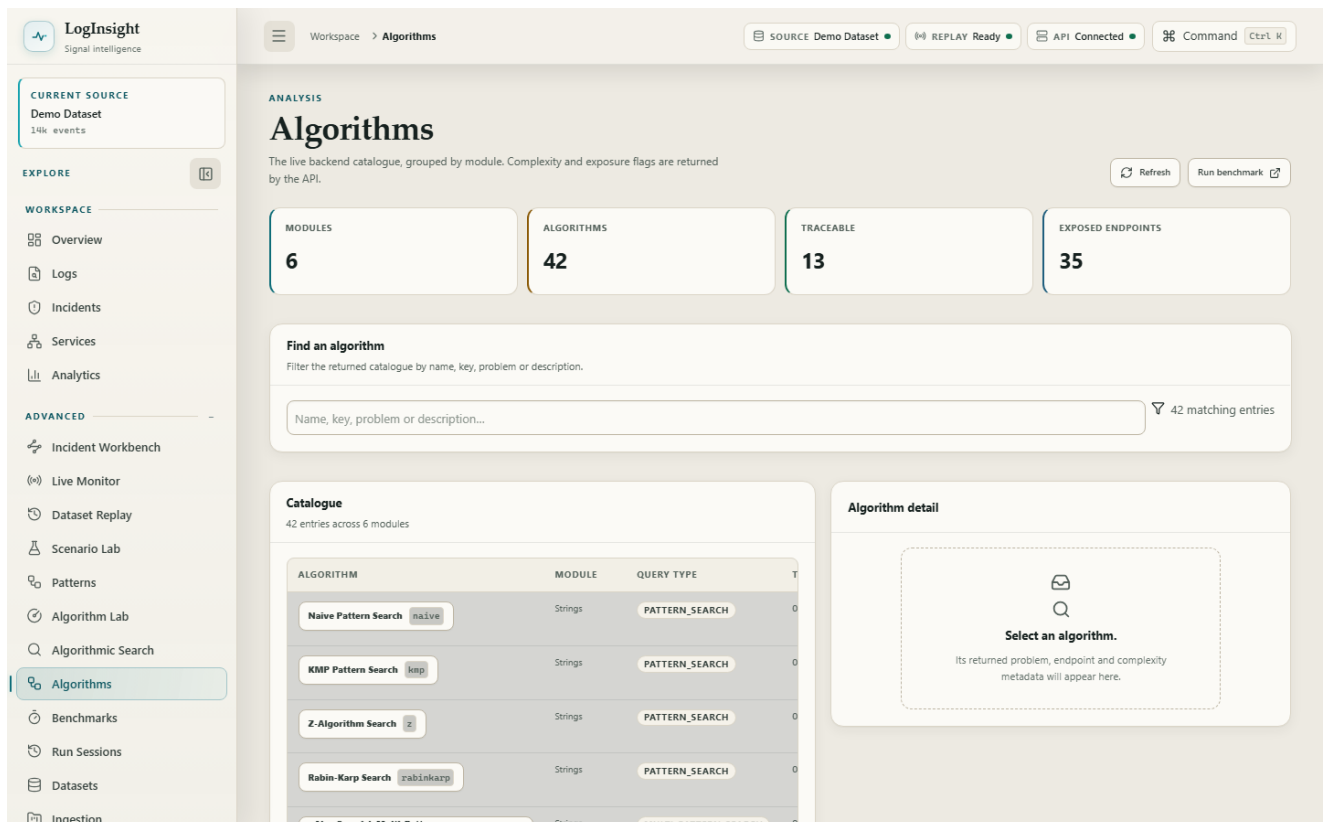
Log explorer with a server-side query and results.



Analytics computed for the selected range.



Service topology with a selected service.



Algorithm catalogue.

LogInsight

Signal intelligence

CURRENT SOURCE

Choose telemetry

Simulation · replay · upload

EXPLORE

WORKSPACE

Overview

Logs

Incidents

Services

Analytics

ADVANCED

API connected

Log intelligence

v0.1.0

Workspace

SOURCE No dataset

REPLAY Standby

API Connected

Command Ctrl K

LOGINSIGHT · OBSERVE

Command center

A workspace for reading operational signals, following service relationships and inspecting evidence.

Refresh

Dataset replay

Live monitor

LogInsight

Current state and next action

Selected-window metrics, observed service relationships and detector windows for the loaded dataset. Every value below is returned by the backend.

Start guided demo

Explore topology

WORKSPACE READY

Awaiting a data source

The backend holds no dataset, so the signal field stays empty rather than showing invented values.

Choose a source

Ingest logs

FROM EVENT TO EVIDENCE

waiting for a backend dataset

DATASET 01

awaiting source

WINDOW 02

load data first

SERVICES 03

graph unavailable

DETECTOR 04

load data first

No telemetry shown until a source is loaded

How data moves

OBSERVED WINDOW

5m

15m

1h

6h

24h

Updated 11 Oct, 08:54:42

YOUR WORKSPACE IS READY

No investigation data yet

Start with a bounded replay, load a bundled sample, or upload logs.

Metrics remain empty until a source is loaded.

Choose a source

Upload logs

Open dataset replay

1 SOURCE

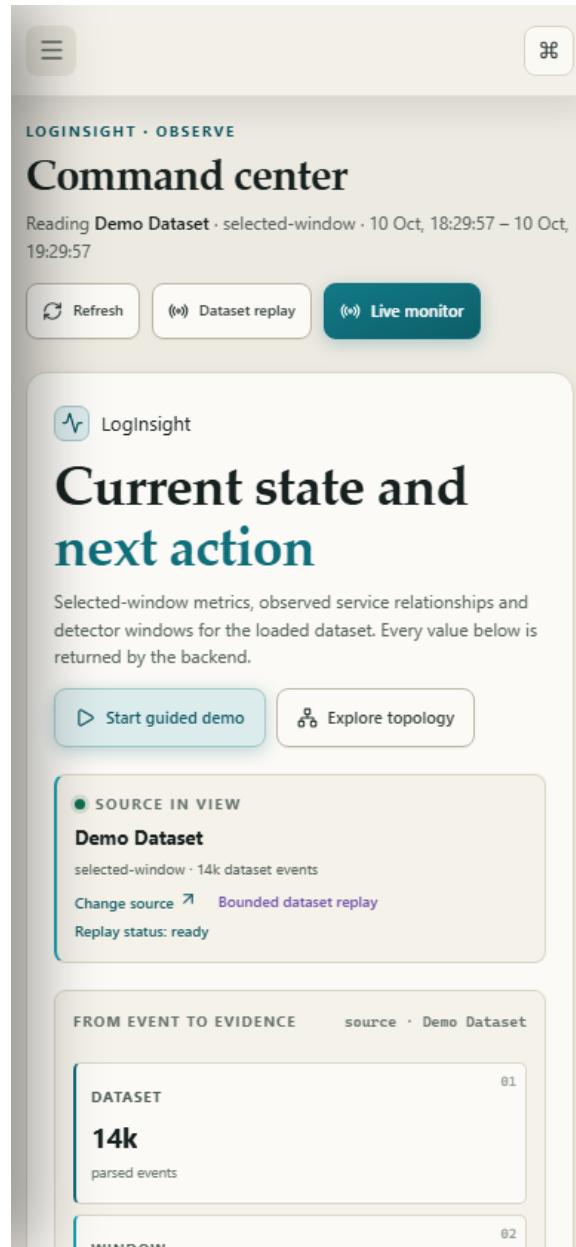
2 OBSERVE

3 INVESTIGATE

No dataset: onboarding rather than fabricated metrics.

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Responsive mobile layout.

10. Known limitations

- In-memory, single-current-dataset state. No database, no durable uploads, no restart persistence.
- No authentication or authorization. This is not a shared multi-user service.
- Dataset Replay is a bounded replay of the loaded dataset, not a live collector.
- Live Monitor is server-generated simulation from a scenario and seed, not captured telemetry.
- Topology edges are request-trail adjacency inferred from requestId co-occurrence, not verified infrastructure.
- Benchmarks are host- and input-dependent measurements, not universal performance claims.
- The 3D topology requires WebGL and falls back to 2D where it is unavailable.
- Generated traffic is not production telemetry and is labelled as such throughout the interface.

11. Future work

- Durable dataset storage so investigations survive a restart.
- Authentication and authorisation, which the current deployment model does not provide.
- Streaming ingestion from a real collector, replacing the generated simulation and bounded replay.
- Distributed algorithm execution for the largest catalogue entries.